RPractical

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2026-11-11

R Markdown

This is an R Markdown document. Markdown is a simple formatting syntax for authoring HTML, PDF, and MS Word documents. For more details on using R Markdown see http://rmarkdown.rstudio.com.

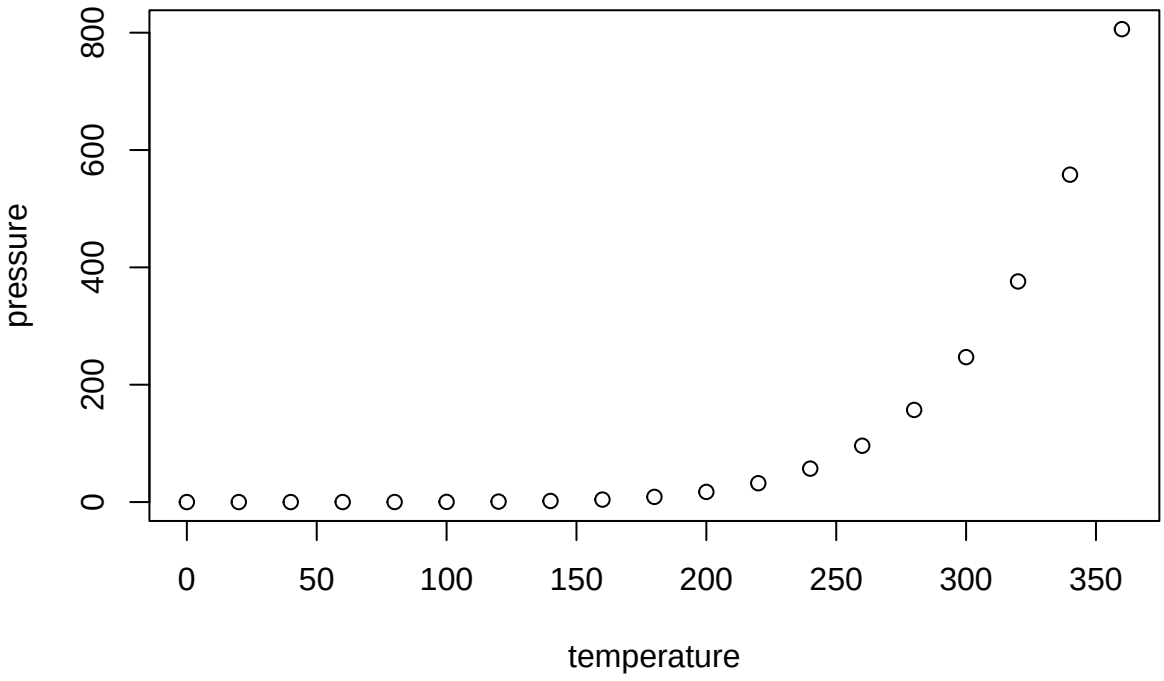
When you click the **Knit** button a document will be generated that includes both content as well as the output of any embedded R code chunks within the document. You can embed an R code chunk like this:

```
summary(cars)
```

```
##      speed           dist
##  Min.   : 4.0   Min.   :  2.00
##  1st Qu.:12.0   1st Qu.: 26.00
##  Median :15.0   Median : 36.00
##  Mean   :15.4   Mean   : 42.98
##  3rd Qu.:19.0   3rd Qu.: 56.00
##  Max.   :25.0   Max.   :120.00
```

Including Plots

You can also embed plots, for example:



Note that the `echo = FALSE` parameter was added to the code chunk to prevent printing of the R code that generated the plot. PART A A1

```
#Question 1
v <- c(seq(5,50, by = 5))
v
```

```
##  [1]  5 10 15 20 25 30 35 40 45 50
```

```
length(v)
```

```
## [1] 10
```

```
sum(v)
```

```
## [1] 275
```

```
mean(v)
```

```
## [1] 27.5
```

A2

```
#Question 2
v <- c(1,2,3)
rep(v, times = 4)
```

```
##  [1] 1 2 3 1 2 3 1 2 3 1 2 3
```

```
rep(v, each = 4)
```

```
##  [1] 1 1 1 1 2 2 2 2 3 3 3 3
```

A3

```
#Question 3
a = 17
b = 5
c = 2
d = 10
e = 144

result1 <- a%%b
result1
```

```
## [1] 2
```

```
result2 <- a%/%b
result2
```

```
## [1] 3
```

```
result3 <- c^d
result3
```

```
## [1] 1024
```

```
result4 <- sqrt(e)
result4
```

```
## [1] 12
```

```
result5 <- round(pi, digits = 3)
result5
```

```
## [1] 3.142
```

A4

```
#Question 4
a <- c(2, 4, 6, 8)
b <- c(1, 3, 5, 7)

total <- a + b
total
```

```
## [1]  3  7 11 15
```

```
product <- a * b
product
```

```
## [1]  2 12 30 56
```

```
logcom <- a > b
logcom
```

```
## [1] TRUE TRUE TRUE TRUE
```

```
dotprod <- prod(a * b)
dotprod
```

```
## [1] 40320
```

A5

```
#Question 5
paste("Student-1", "Student-2", "Student-3")
```

```
## [1] "Student-1 Student-2 Student-3"
```

```
mode(42)
```

```
## [1] "numeric"
```

```
mode(42L)
```

```
## [1] "numeric"
```

```
mode("42")
```

```
## [1] "character"
```

```
mode(41i)
```

```
## [1] "complex"
```

```
mode(TRUE)
```

```
## [1] "logical"
```

PART B B1

```
id <-  c(1, 2, 3, 4, 5, 6, 7, 8)
name <-  c("Alice", "Bruno", "Chen", "Dana", "Elif", "Farid", "Grace", "Hiro")
sex <- c("F", "M", "M", "F", "F", "M", "F", "M")
section <- c("A", "B", "A", "C", "B", "C", "A", "B")
midterm <- c(78, 85, 92, 67, 74, 88, 95, 59)
final <- c(82, 79, 96, 71, 80, 91, 93, 65)
attendance <- c(95, 88, 100, 72, 90, 85, 98, 60)
```

```
students <- data.frame( id = id, sex = sex, section = section, midterm = midterm, final = final, attenda
students
```

```
##   id sex section midterm final attendance
## 1  1   F       A      78    82         95
## 2  2   M       B      85    79         88
## 3  3   M       A      92    96        100
## 4  4   F       C      67    71         72
## 5  5   F       B      74    80         90
## 6  6   M       C      88    91         85
## 7  7   F       A      95    93         98
## 8  8   M       B      59    65         60
```

B2

```
str(students)
```

```
## 'data.frame':    8 obs. of  6 variables:
##  $ id        : num  1 2 3 4 5 6 7 8
##  $ sex       : chr  "F" "M" "M" "F" ...
##  $ section   : chr  "A" "B" "A" "C" ...
##  $ midterm   : num  78 85 92 67 74 88 95 59
##  $ final     : num  82 79 96 71 80 91 93 65
##  $ attendance: num  95 88 100 72 90 85 98 60
```

```
dim(students)
```

```
## [1] 8 6
```

```
names(students)
```

```
## [1] "id"         "sex"        "section"    "midterm"    "final"
## [6] "attendance"
```

```
summary(students)
```

```
##        id              sex         section     midterm          final
##  Min.   :1.00   Length   :8   Length   :8   Min.   :59.00   Min.   :65.00
##  1st Qu.:2.75   N.unique :2   N.unique :3   1st Qu.:72.25   1st Qu.:77.00
##  Median :4.50   N.blank  :0   N.blank  :0   Median :81.50   Median :81.00
##  Mean   :4.50   Min.nchar:1   Min.nchar:1   Mean   :79.75   Mean   :82.12
##  3rd Qu.:6.25   Max.nchar:1   Max.nchar:1   3rd Qu.:89.00   3rd Qu.:91.50
##  Max.   :8.00                               Max.   :95.00   Max.   :96.00
##    attendance
##  Min.   : 60.00
##  1st Qu.: 81.75
##  Median : 89.00
##  Mean   : 86.00
##  3rd Qu.: 95.75
##  Max.   :100.00
```

B3

```
students$average <- round((students$midterm + students$final) / 2, 1)
students
```

```
##   id sex section midterm final attendance average
## 1  1   F       A      78    82         95    80.0
## 2  2   M       B      85    79         88    82.0
```

```
## 3  3   M       A      92    96        100    94.0
## 4  4   F       C      67    71         72    69.0
## 5  5   F       B      74    80         90    77.0
## 6  6   M       C      88    91         85    89.5
## 7  7   F       A      95    93         98    94.0
## 8  8   M       B      59    65         60    62.0
```

B4

```
students$passed <- students$average >= 70 & students$attendance >= 75
students
```

```
##   id sex section midterm final attendance average passed
## 1  1   F       A      78    82         95    80.0   TRUE
## 2  2   M       B      85    79         88    82.0   TRUE
## 3  3   M       A      92    96        100    94.0   TRUE
## 4  4   F       C      67    71         72    69.0  FALSE
## 5  5   F       B      74    80         90    77.0   TRUE
## 6  6   M       C      88    91         85    89.5   TRUE
## 7  7   F       A      95    93         98    94.0   TRUE
## 8  8   M       B      59    65         60    62.0  FALSE
```

B5

```
students$standing <- ifelse(students$average >= 90, "Honors",
                     ifelse(students$average >= 75, "Good", "Needs Support"))
students
```

```
##   id sex section midterm final attendance average passed      standing
## 1  1   F       A      78    82         95    80.0   TRUE          Good
## 2  2   M       B      85    79         88    82.0   TRUE          Good
## 3  3   M       A      92    96        100    94.0   TRUE        Honors
## 4  4   F       C      67    71         72    69.0  FALSE Needs Support
## 5  5   F       B      74    80         90    77.0   TRUE          Good
## 6  6   M       C      88    91         85    89.5   TRUE          Good
## 7  7   F       A      95    93         98    94.0   TRUE        Honors
## 8  8   M       B      59    65         60    62.0  FALSE Needs Support
```

B6

```
classify <- function(score) {
  if (score >= 85) {
    "Distinction"
  } else if (score >= 70) {
    "Merit"
  } else if (score >= 50) {
    "Pass"
  } else {
    "Fail"
  }
}

result <- sapply(c(90, 73, 55, 40), classify)
print(result)
```

```
## [1] "Distinction" "Merit"       "Pass"        "Fail"
```

PART C C1